# LAB 4: SIMULATION OF ANGULAR VELOCITY CONTROL FOR A DC MOTOR

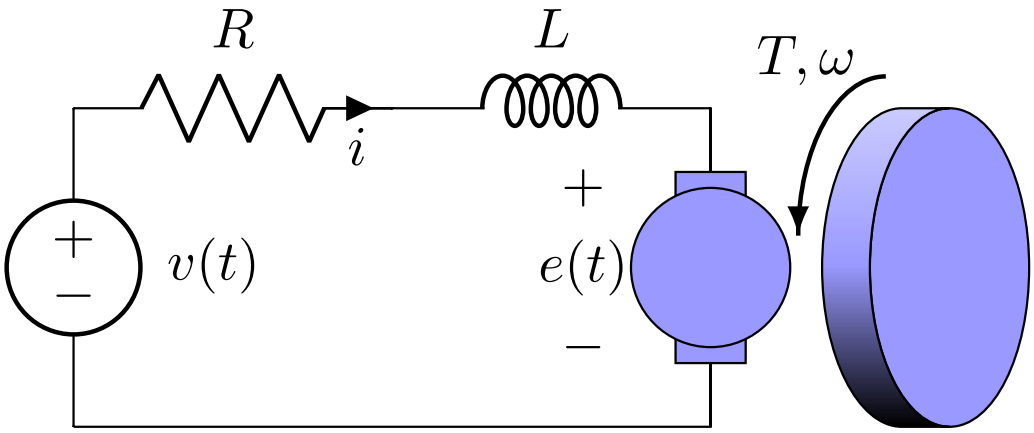
```
%--------------------------------------
% Fill in your student Name and ID.
Students.Name = '';
Students.ID = '';
%--------------------------------------
```

## 0. Objectives

In this lab, we will run different simulation experiments to track a desired reference signal with a simulated DC Motor. We will first run an open-loop experiment, then we will use state feedback to control the angular velocity of DC motor.

## 1. A State-Space Representation of DC Motor

A DC motor is a device that converts armature current into mechanical torque. A schematic of a DC motor is shown in the following




We can model the DC motor using the armature current $i$ and the angular speed $\omega$. The dynamics of the DC motor are given by the following two coupled differential equations:

$$J\frac{d\omega}{dt} + b\omega = K_t i$$

$$L\frac{di}{dt} + Ri = v - K_e\omega$$

where the parameters are defined as follow

```
Param.Kt = 0.034; % Torque proportionality constant (Kg m2 /s2/A) (N.m/A)
Param.Ke = 0.034; % Back electromotive proportionality constant (V/rad/s)
Param.L = 0.018; % Electric inductance (H)
Param.R = 8.4; % Terminal resistance (Ohm)
Param.J = 2e-05; % Rotor moment of inertia (kg.m2)
Param.b = 1e-06; % Rotor viscous friction constant (Kg.m2/s) (N.m.s)
```

From these differential equations we can represent the DC Motor model by the following state-space representation

$$\frac{d}{dt}x(t) = \begin{bmatrix} -\frac{b}{J} & \frac{K_t}{J} \\ -\frac{K_e}{L} & -\frac{R}{L} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} v(t)$$

where $v(t)$ is the voltage and the state vector contains the angular velocity $\omega(t)$ and the current $i(t)$

$$x(t) = \begin{bmatrix} \omega(t) \\ i(t) \end{bmatrix}$$

## *Question 1 (1 mark)*

Write the state-space representation of the DC Motor (Only A and B matrices).

```
A
B
```

## 2. Desired specifications

The objective of this lab is to track a reference signal $\omega_{ref}$ while satisfying the following requirements:

- Overshoot less than 10%.
- 5%-settling time less than 0.2 seconds.
- Steady state gain of $1 \pm 5\%$.

The Reference angular velocity is defined by the following:

$$\omega_{ref}(t) = \begin{cases} 0, & 0 \leqslant t < 1 \\ 50, & t \geqslant 1 \end{cases}$$

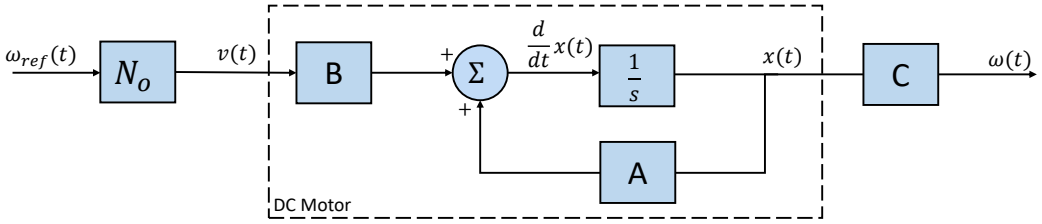
In **MATLAB** we define $\omega_{ref}$ by:

```
T = 10; % Simulation duration
dt = 0.01; % Simulation step time
t = (0:dt:T)'; % Time array
TStart = 1; % Step start
wRef = zeros(size(t));
wRef(t >= TStart) = 50; % rad/s

figure;
clf;
plot(t, wRef, 'LineWidth', 2.0)
legend('Reference angular velocity (rad/s)');
xlim([0, T])
ylim([0, 80])
title(sprintf('Student ID %s - Reference angular velocity', Students.ID));
grid on
```

## 3. Open-loop Simulation Experiment

The state-space open-loop system is presented below.




Note that if we think of the angular velocity $\omega(t)$ as the output of this system, then $C = [1 \quad 0]$ and

$$\omega(t) = Cx(t)$$

In this case, the transfer function from the voltage $v(t)$ to the angular velocity $\omega(t)$ is given by

$$G_o(s) = C(sI - A)^{-1}B$$

In **MATLAB**, we can find the transfer function of a state-space model using the function **ss** and **tf**, as shown below.

```
C = [1, 0];
D = 0;
Go = ss(A, B, C, D)
tf(Go)
```

## *Question 2 (1 mark)*

Recall that the DC gain can be obtained from a Laplace transfer function, by setting the laplace variable $s$ to zero. From the state-space model representation $(A, B, C, D)$, calculate the DC gain.

```
dcGainOL
```

Observe that if the open-loop system is a stable system. The DC Motor will be responded to a reference step signal without using a controller. However, the response will be determined by the eigenvalues of the open-loop $A$ matrix (or equivalently, the poles of the open-loop transfer function).

## *Question 3 (1 mark)*

Determine the poles of the open-loop transfer function from the state-space matrix $A$ (Hint: **`help eig`**). Is the system stable? (Justify your answer.)

```
p1
p2
```

```
% Write your answer here.
```

Following a fixed input $v_0$ the angular velocity $\omega$ will attain a steady angular velocity $\omega_\infty = DC_{gain} v_0$.

We can simulate the response of an LTI system to an arbitrary input using the **`lsim`** function.

## *Question 4 (0.5 mark)*

Complete the following code to simulate the response of the system to a 1 volt input signal, and plot the results.

```
v = ones(size(t)); % create a signal with one volt.
w = lsim(Go, v, t);

% Plot results
figure;
clf;

% [Answer here] plot the angular velocity w

legend('Angular velocity (rad/s)', ...
    'location', 'best');
xlim([0, T])
ylim([0, 40])
title(sprintf('Student ID %s - Simulation of 1 volt step', Students.ID));
grid on
```

In open-loop we can generate the voltage $v$ from $\omega_{ref}$ through the gain $N_o$.

$$v(t) = N_o \omega_{ref}(t)$$

## *Question 5 (1 mark)*

Find the value of $N_o$ such the final value of the angular velocity output $\omega$ is the same as $\omega_{ref}$ (i.e. 50 rad/s)

```
No


% The voltage is generated from wRef and No
v = No * wRef;
% [Answer here] Simulate the output angular velocity using lsim

w


% Plot results
figure;
clf;

% [Answer here] Superpose the angular velocity w and the reference wRef

legend('Angular velocity (rad/s)', ...
    'Reference angular velocity (rad/s)', ...
    'location', 'best');
xlim([0, T])
ylim([0, 80])
title(sprintf('Student ID %s - Open-loop tracking Simulation', Students.ID));
grid on
```

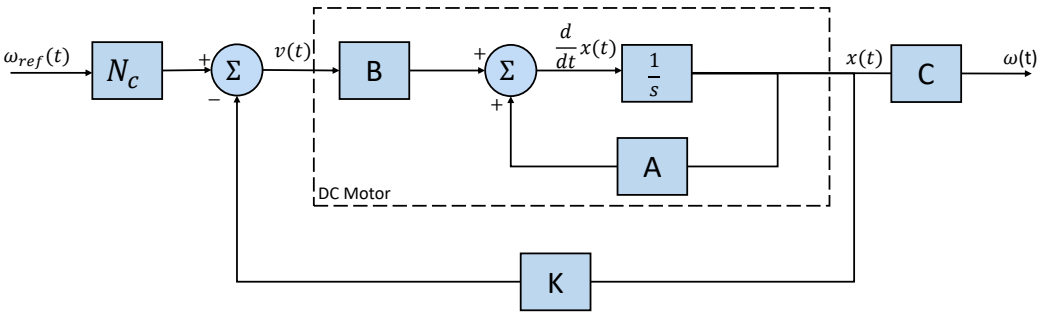
## *Question 6 (0.5 marks)*

Based on the plot above, determine if all the requirements are satisfied. Justify your answer.

Answer here

# 4. Closed-Loop Simulation Experiment

The closed-loop system is presented in the following figure.

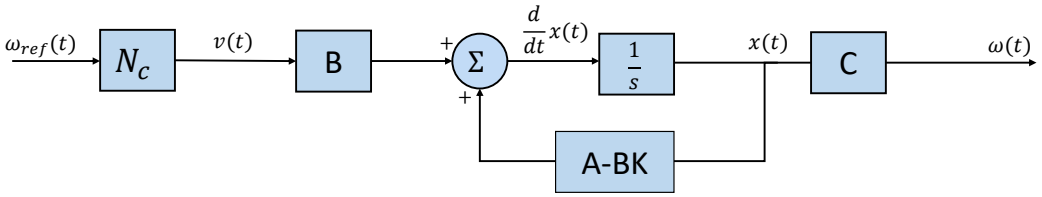



The full-state feedback control law is defined as follow

$$v(t) = N_c \omega_{ref} - Kx(t)$$

where gain $K$ is a 1 by 2 matrix and the gain $N_c$ is forward gain that will ensure achieving $\omega_{ref}$ at steady state.

Actually, this diagram is equivalent to an open-loop diagram but with a different state transition matrix $A_{cl} = A - BK$




Doing a state feedback with gain $K$ is equivalent to changing the system dynamics from $A$ to $A - BK$ . Recall that the eigenvalues of a state transition matrix $A$ are the poles of the system.

We want to choose $K$ to specify the new poles of the closed-loop system $A - BK$ in a way that satisfies the desired specifications

- Overshoot less than 10%.
- 5%-settling time less than 0.2 seconds.

Then we will choose $N_c$ similarly to the open-loop section to satisfy the steady state requirements:

- Steady state gain of 1 $\pm$ 5%.

## *Question 7 (2 marks)*

Calculate the desired poles that satisfies the specifications (Use formula sheet for transient response of 2nd order system in myCourses if needed):

```
p1d
p2d
```

Instead of manual calculation of the value of the gain $K$ that place the poles of $A - BK$ at [p1d, p2d], we will use the **`place`** command to calculate the gain $K$

```
K = place(A, B, [p1d, p2d]);
```

## *Question 8 (1 mark)*

The closed-loop transfer function is

$$G_c(s) = C(sI - (A - BK))^{-1}B$$

1. What is the value of the DC gain of the closed-loop system.
2. What is the value $N_c$ needed to compensate for the DC gain of the closed-loop system.

```
dcGainCL
```

```
Nc
```

## *Question 9 (1 mark)*

Complete the following to simulate angular velocity control with state feedback that tracks the desired $\omega_{ref}$.

[Answer here] Define the closed-loop system using ss

```
Gc


% [Answer here] Simulate the output angular velocity using lsim

w


% We also simulate the voltage for visualization
v = -lsim(ss(Gc.A, Gc.B, eye(2), Gc.D), Nc*wRef, t) * K' + Nc * wRef;

figure;
clf;

subplot(2, 1, 1)
% [Answer here] Superpose angular velocity w with the angular velocity
% reference

legend('angular velocity (rad/s)', ...
    'Reference angular velocity (rad/s)', ...
    'location', 'best');
xlim([0, T])
ylim([0, 80])
grid on
title(sprintf('Student ID %s - Closed-loop tracking Simulation', Students.ID));

subplot(2, 1, 2)
% [Answer here] Plot the voltage

legend('voltage (V)', 'location', 'best');
xlim([0, T])
grid on
```

## *Question 10 (1 marks)*

Based on the plot above, determine if all the requirements are satisfied. Justify your answer.

```
% Answer here
```